

Sky & Space: Class 8

A year of naked-eye astronomy for class 8: read the night sky with charts and handspans, understand light and the instruments that catch it, then follow that light out past the Sun to the Milky Way and the galaxies beyond.

12	32	15 h 20 min	2 h 40 min	School: Middle (6–8)
SESSIONS	CONCEPTS TAUGHT	TOTAL	OUTDOORS	AUDIENCE

Moon: start when the moon is **2.75–6.5 days old** (🌑 waxing crescent → 🌒 first quarter)

ASSUMED ALREADY KNOWN

Phases of the Moon

Seasons

Gnomon & Local Noon

Earth is a Sphere

Tropics & Polar Circles

Latitude & Longitude

Angles & the Protractor

The Solar System Family

Scale of the Solar System

Big Numbers & Powers of Ten

Speed, Distance & Time

Reflection & Refraction

Rainbows & Colours

How We See

The Sun Powers Earth

Closed under prerequisites: 29 concepts in total are taken as prior knowledge.

1

Stars are not all the same

CLASSROOM · 1 H 20 MIN

KIT

Projector · Star chart / planisphere

L1

Stars Differ

Stars are not identical dots: they differ in brightness and in colour — steely blue Vega, golden Arcturus, red Betelgeuse. These differences are real physics, visible to the naked eye.

L1

Constellations

Star patterns used as sky landmarks — 88 official constellations today, with older patterns in every culture. They are direction labels, not physical groups.

L2

Star Names & Bayer Letters

Bright stars carry ancient proper names (many Arabic: Aldebaran, Deneb); astronomers label stars α , β , γ by brightness within each constellation. Two naming systems, one sky.

TRAINER NOTE

Open the year by asking who has actually looked at the night sky for ten unbroken minutes — in a city class, almost nobody. Show a long-exposure sky photo and ask what differs star to star: brightness, colour, and nothing else the eye can get. Constellations must land as human inventions, not physical objects — draw Orion's stars, then reveal their wildly different distances. Star names are the hook for homework: send them out tonight to find one named star and report back. That homework is the course.'

2 Handspans and star charts

INDOOR LAB · 1 H 20 MIN

KIT Star chart / planisphere · Paper, pencils, protractor

L2 Angles on the Sky

Sky positions and sizes are measured in angles: your fist at arm's length spans about 10° , the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

L2 Star Charts & Planetarium Apps

Reading a sky map: orientation, scale, magnitudes, and matching chart to sky. A planisphere or app turns the catalogue sky into 'what's above me right now'.

L2 Observing Logs & Sketching

Date, time, sky conditions, instrument, and what you actually saw — a log turns stray sessions into a record of growth, and sketching trains the eye to see more.

TRAINER NOTE Calibrate every student's own hand: fist $\approx 10^\circ$, span $\approx 20^\circ$, against a wall chart they measure with a protractor first. Then hand out the planisphere and make them fail at it before you explain it — set the wrong date, find the 'error', fix it. Start the observing log [HERE](#), this week, with a naked-eye entry (moon position against their fist-measured landmark). A log started in week 2 gets kept; a log started before the observing night does not.

3 The wheeling sky and the star that stands still

CLASSROOM · 1 H 20 MIN

KIT Projector · Star chart / planisphere

L2 Diurnal Motion

The whole starry sky wheels westward through the night, circling a fixed point — the celestial pole. This is Earth's rotation seen from the inside.

L2 The Pole Star

Polaris (Dhruva) sits nearly at the north celestial pole, so it barely moves and always marks north. Its altitude above the horizon equals your latitude.

L2 Indian Star & Constellation Names

Dhruva (Polaris), Saptarshi (the Big Dipper), Arundhati–Vasishtha (the Mizar–Alcor pair), Rohini (Aldebaran), Chitra (Spica): the Indian sky map atop the same stars.

TRAINER NOTE Play a star-trail time-lapse before any diagram: the sky visibly turns around one point. Every class contains the misconception that the pole star is the brightest star — put it to a vote, then show Dhruva sitting at magnitude 2, forty-eighth in line. The Dhruva–Saptarshi story is not decoration: 'the seven rishis walk around the fixed prince' IS diurnal motion, taught by grandmothers for centuries. Show how the Saptarshi pointer stars find Dhruva; that trick gets tested on the observing night.

4 Wanderers and the road they keep

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L2 Planets: the Wanderers

A few bright 'stars' drift against the fixed constellations from week to week — the planets. They shine steadily while true stars twinkle.

L2 Seasonal Skies

The night sky changes with the calendar — Orion rules winter evenings, Scorpius the monsoon sky — because Earth's orbit points our nights toward different stars each season.

L2 Ecliptic & Zodiac

The Sun's yearly path against the stars is the ecliptic; the belt of constellations it crosses is the zodiac. The Moon and planets always keep near this line.

TRAINER NOTE 'Planet' must stop meaning 'a ball in a diagram' and start meaning 'a bright star that cheats' — show the same constellation photographed months apart with a planet moved. Seasonal skies answer a question someone will have asked by now: why can't we see Orion in May? Walk the classroom model: lamp Sun in the centre, zodiac posters on the walls, students as Earth — night side faces different walls in different months. The ecliptic then falls out for free: the wanderers keep to the same twelve posters. Do NOT explain retrograde loops if asked — write the question on the board and promise it to class 9. A kept promise there is worth more than a rushed answer here.

5 Noon: measuring the Earth with a stick

SOLAR SESSION · 1 H 20 MIN · DAY 30

SKY needs Any moon · The Sun

KIT Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor

L3 Measuring the Earth

Eratosthenes sized the Earth around 240 BCE with two sticks and a noon shadow angle. Two schools at different latitudes can repeat it today over WhatsApp.

L2 Zero Shadow Day

Twice a year, everywhere between the tropics, the noon Sun stands exactly overhead and shadows vanish for a moment. India's favourite mass astronomy experiment — date depends on your latitude.

TRAINER NOTE The flagship. A vertical stick at local noon gives the sun's altitude; a partner school a few hundred km north or south (arrange this in week 1, not week 9) gives a second angle and the class computes the circumference of the Earth to within a few percent. Let them do the arithmetic — the wrong answers matter. Zero-shadow day is the local payoff: compute your town's date from its latitude, then check whether your latitude even gets one. If it falls in term time, schedule the five-minute stand-in-the-yard moment; a shadowless student is a photograph they keep.

6 Light: the only messenger

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L2 Waves

Ripples on water and sound in air both carry energy as waves, with a wavelength and a pitch. Light is a wave too — its 'pitch' is colour, and motion can stretch or squeeze it.

L2 Light & Lookback Time

Light is fast but finite — 8 minutes from the Sun, 4 years from the nearest star. Every telescope is a time machine: we see objects as they were, not as they are.

L3 Spectrum & Colour

White light is a mixture; a prism fans it into the spectrum, red to violet, each colour a different wavelength. Colour is the first physical measurement of starlight.

TRAINER NOTE Everything astronomy knows arrives as light, so spend a period on the messenger itself. Light having a speed is the shock: 1.3 seconds from the Moon, 8 minutes from the Sun — so the Sun you see is 8 minutes old, and someone will ask the right next question about the stars; hold that thread, session 11 pays it. A prism or a water-tray rainbow splits sunlight live — they met the rainbow in class 7, but now frame the spectrum as information carried by the light, not just decoration. Waves are the vocabulary for that: wavelength = colour is the one fact to nail down.

7 Bending light, and the eye at night

INDOOR LAB · 1 H 20 MIN

KIT Paper, pencils, protractor · Projector

L2 The Eye as a Detector

Cone cells see colour but need light; rod cells see faint things only in grey. That's why nebulae look ghostly-grey in the eyepiece yet glow red and blue in photographs.

L2 Dark Adaptation & Averted Vision

Eyes take 20–30 minutes to reach full night sensitivity, and one white-light glance resets them. Looking slightly beside a faint object uses your more sensitive off-centre retina.

L3 Lenses, Mirrors & Focal Length

A curved lens or mirror gathers parallel starlight to a focus; the focal length sets image scale. Real images on a screen — this is where 'optics' becomes 'instrument'.

TRAINER NOTE Every pair of hands gets a lens: magnifier + sunlit window = an image of the world upside down on paper. They know refraction from class 7; the new idea is that a curve ORGANISES the bending into an image — that is the whole secret of every telescope they will ever meet. Then the instrument they already own: the eye, with its iris-aperture and its slow chemical night mode. Run the lights-off drill — blackout the lab for ten minutes and have them time how long before they can read a dim card. That number (and why one flash of white light resets it) is the discipline the observing night runs on. Introduce the red torch here and make its logic explicit.

8 Binoculars, telescopes, and what Galileo saw

INDOOR LAB · 1 H 20 MIN

KIT Binoculars · Dobsonian reflector · Eyepiece set

L2 Binocular Astronomy

A 7×50 binocular — 7× magnification, 50 mm objectives — shows Moon craters, Jupiter's moons and hundreds of clusters. The best first instrument, and many observers' favourite forever.

L2 First Light: Using a Telescope

Align the finder in daylight, start with the lowest power, centre on something bright, focus slowly. Ten minutes of drill separates a magical first night from a frustrating one.

L3 Galilean Moons

Jupiter's four big moons shuffle visibly from night to night — a miniature solar system in any small telescope, and Galileo's proof that not everything orbits Earth.

TRAINER NOTE Daytime handling drill, no sky needed: focus binoculars on a distant signboard, learn to brace elbows or rest on a wall — hand-shake is the beginner-killer and it is fixable in one afternoon. Then the dobsonian: name the parts, find, centre and focus on a terrestrial target, and rehearse the queue discipline (don't touch the tube, ask for the focuser). End with Galileo's 1610 notebook pages: four dots that moved night after night, and what that did to the idea that everything orbits Earth. The class will re-perform that observation with their own eyes on the observing night if Jupiter is up — check whatsup.html before promising.

9 Evening under the crescent

OBSERVING NIGHT · 1 H 20 MIN · DAY 60

SKY needs **Thin crescent** · The Moon, Constellations, Bright stars, Milky Way

KIT Binoculars · Dobsonian reflector · Eyepiece set · Star chart / planisphere · Red torch

L2 Light Pollution

Artificial skyglow erases faint stars; the Bortle scale grades sky darkness from 1 (pristine) to 9 (inner city). Shielded, warm, dimmed lighting brings stars back.

L2 The Milky Way Band

From a dark site a faint luminous band arches across the sky — the combined light of billions of stars in our galaxy's disc, seen from inside it.

L2 Earthshine

The ghostly glow filling the dark part of a crescent Moon is sunlight reflected off Earth — 'the old Moon in the new Moon's arms'. You are seeing our planet light another world.

L2 Lunar Observing

Craters, maria, mountain shadows — the Moon is best along the terminator, where low sunlight throws relief into drama. Full moon is actually the flattest, worst time.

TRAINER NOTE The night the year points at. A crescent is the right moon three times over: the terminator throws crater shadows a full moon flattens, earthshine is visible ('the old moon in the new moon's arms' — make them explain whose light that is, it is the best viva question of the year), and it sets early enough to leave real darkness. Run the moon first through the dobsonian while twilight fades and eyes adapt. Then charts out, red torches only: find Dhruva by the Saptarshi pointers, log two constellations. If the site is dark enough for the Milky Way band, end there; if it is not, THAT is the light-pollution lesson, taught by the sky itself — count stars in the Saptarshi bowl and compare with a dark-site photo. Every student leaves with a log entry written at the eyepiece, not at home from memory.

10 The Sun is a star

CLASSROOM · 40 MIN

KIT Projector

L2 The Sun is a Star

Our Sun is an ordinary star seen close-up — a million-km fusion furnace. Every night-sky star is a sun; many have their own planets.

TRAINER NOTE One idea, one period, and it is the pivot of the whole course: the Sun is a star seen close up, and every star is a sun seen from very far away. Run the thought experiment in both directions — drag the Sun away until it is a dot (how far until it looks like Sirius?), drag Sirius in until it is a disc. If the class believes this one sentence, the remaining two sessions are consequences; if they don't, the galaxy is just vocabulary. Take the full forty minutes and take objections seriously.

11 Light years: putting a ruler on the stars

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor · Measuring tape

L3 Nebulae & Star Clusters

Glowing gas clouds where stars are born; open clusters of young siblings; ancient globular swarms of hundreds of thousands. The Milky Way's visible ecology.

L3 Light Years & Cosmic Scale

The nearest star is 4.2 light years away — 7,000× farther than Neptune. Grasping the jump from solar-system scale to interstellar scale is a rite of passage.

TRAINER NOTE Cash in session 6's IOU: light from Alpha Centauri left 4.3 years ago, so a light year is a RULER, not a duration — the most common class 8 error is filing it under time, and the name is to blame; say so. Scale model in the corridor: Sun a marble at one end, Earth 2.5 cm away, and the nearest star... 700 km away, another town. Let them compute that last number themselves; disbelief is the correct reaction. Then what the darkness between holds: run the Pleiades (which they can find with binoculars) and the Orion Nebula as the two live examples — star cluster and star nursery, both logged targets for their own next dark evening.

12 Our galaxy, and the others

CLASSROOM · 1 H 20 MIN

KIT Projector

L4 Structure of the Milky Way

A barred spiral of 100–400 billion stars, 100,000 ly across, with us in a quiet arm 27,000 ly out. The naked-eye band is our inside view of the disc.

L4 Galaxies

Islands of billions of stars — spirals, ellipticals, irregulars — separated by millions of light years. Andromeda, our giant neighbour, is faintly visible to the naked eye.

TRAINER NOTE Close the loop with the observing night: that faint band was our own galaxy viewed from inside — a disc seen edge-on from a seat two-thirds of the way out. The hard cognitive step is being inside the thing being described; the classroom-as-disc walkthrough (look along the rows = many heads = the band, look up = few) does more than any artist's impression, every one of which should be labelled 'not a photograph — nobody has ever left to take one'. Then the kicker: the Andromeda smudge, visible to their naked eye on a dark night, is another entire Milky Way — the farthest thing an unaided human eye can see, and light 2.5 million years old landing on their retina. End the year there, with the size of everything, and hand the observing logs back with a comment each.

FULL KIT LIST

Binoculars	sessions 8, 9
Dobsonian reflector	sessions 8, 9
Eyepiece set	sessions 8, 9
Gnomon / shadow stick (<i>consumable</i>)	sessions 5
Measuring tape	sessions 5, 11
Projector	sessions 1, 3, 4, 6, 7, 10, 11, 12
Red torch	sessions 9
Star chart / planisphere	sessions 1, 2, 3, 9
Paper, pencils, protractor (<i>consumable</i>)	sessions 2, 4, 5, 6, 7, 11